Question	Answer	Marks	Partial Marks
3(a)(iii)	1706 cao nfww	**3**	**B2** for 1705 to 1706.0… or 1710 or **M1** for $\left(1+\frac{30.48}{100}\right)x = 2226$ oe or better If 0 or M1 scored, **SC1** for rounding *their* decimal answer seen to nearest integer
3(b)	3897	**5**	**B1** for $a = 2000$ **M2** for [b =] $\sqrt[3]{\frac{2662}{2000}}$ or **M1** for $2662 = 2000b^3$ **M1** for *their* $2000 \times \left(\sqrt[3]{\frac{2662}{their\,2000}}\right)^7$ or for *their* $a \times$ (*their* b)^7 provided *their* a and *their* b are clearly identified in the working If 0 or M1 scored, **SC1** for rounding *their* decimal answer seen to nearest integer.
4(a)	$\frac{(12-2)\times 180}{12}$ [= 150] oe or $180 - \frac{360}{12}$ [= 150]	**1**	Accept $\frac{(2\times 12-4)\times 90}{12}$ [= 150]
4(b)(i)	$\frac{3}{\cos 75}$ oe or $\frac{6\sin 75}{\sin 30}$	**M2**	**M1** for $\frac{3}{AO} = \cos 75$ oe or $\frac{r}{\sin 75} = \frac{6}{\sin 30}$
	11.59…	**A1**	
4(b)(ii)(a)	72.8 or 72.9 or 72.82 to 72.89…	**2**	**M1** for $2 \times \pi \times 11.6$
4(b)(ii)(b)	12.1 or 12.06 to 12.08	**2**	**M1** for [6 +] *their* **(b)(ii)(a)** ÷ 12 oe
4(c)	806 or 807 or 805.9 to 807.4	**3**	**B2** for 402.9… to 403.7 OR **M2** for $\frac{1}{2}\times 6\times 11.6\times \sin 75\times 12\times 2$ oe or **M1** for $\frac{1}{2}\times 6\times 11.6\times \sin 75\,[\times k]$ oe
5(a)(i)	$20 < t \leqslant 35$	**1**	

Question	Answer	Marks	Partial Marks
5(a)(ii)	28 nfww	**4**	**M1** for midpoints **soi** **M1** for use of $\sum fm$ with m in correct interval including both boundaries **M1** (dep on 2nd M1) for $\sum fm \div 80$
5(b)(i)	$\frac{7}{8}$ cao	**2**	**M1** for $\frac{18+28+24}{80}$ oe
5(b)(ii)	$\frac{25}{126}$ oe	**3**	**M2** for [2 ×] $\left(\frac{3}{28}\times\frac{25}{27}\right)$ or [2 ×] $\left(\frac{25}{28}\times\frac{3}{27}\right)$ oe or **M1** for either $\frac{3}{28}$ or $\frac{25}{27}$ or $\frac{25}{28}$ or $\frac{3}{27}$ If 0 scored, **SC1** for answer $\frac{75}{392}$ oe
5(c)(i)	28 and 56	**1**	
5(c)(ii)	Correct diagram	**3**	**B1FT** *their* **(c)(i)** for plots at 5 correct heights **B1** for 5 plots at upper ends of intervals on correct vertical line **B1FT** (dep on at least B1) for increasing curve or polygon through 5 points After 0 scored, **SC1FT** for 4 correct points plotted
5(c)(iii)	Strict FT *their* reading at 80th percentile for an increasing curve/polygon	**2**	**B1** for 64 written or a mark at cf = 64 on graph or a mark on curve at (t, 64)
5(c)(iv)	Correct integer reading at $t = 45$	**M1**	**FT** *their* cf graph for all three marks
	$\frac{80-(\textit{their}\text{ reading at } t=45)}{80}$ [× 100] or $\frac{(\textit{their}\text{ reading at } t=45)}{80}\times 100$	**M1**	
	Percentage consistent with *their* reading	**A1**	If no working shown then SC1 for a correct percentage that follows from a correct reading from *their* graph.
6(a)	$5b - 2a$ final answer	**2**	**B1** for $5b$ or $-2a$ in final answer or for $5b - 2a$ seen
6(b)	$6x - 23$ final answer nfww	**2**	**M1** for $4x - 20$ or $-3 + 2x$